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TEAM NEXUS RESOLVE

The Nexus Dome

A Secure & Intelligent Smart Home Ecosystem

Predict. Protect. Automate.

Presented by
Team Nexus Resolve
IoT Internship Project • 2026
Innovation & Incubation Hub
MNNIT Foundation, Prayagraj

Redefining Smart Living.

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Problem Statement & Motivation

Current Challenges

-  **High Energy Consumption**
Unnecessary power usage due to manual operation.
-  **No Centralized Monitoring**
Users cannot monitor home status remotely.
-  **Delayed Emergency Detection**
Late response to gas leaks and fire hazards.
-  **Manual Appliance Control**
Daily appliance management depends on human intervention.
-  **No Predictive Intelligence**
Systems react after problems instead of anticipating them.



Traditional Home

High Energy Usage

Safety Risks

Manual Control

No Intelligence

Poor User Experience

Existing smart home systems primarily automate devices but lack centralized intelligence, predictive monitoring, and proactive decision-making for safer and more efficient living.



Our Solution

The Nexus Dome

A Secure & Intelligent Smart Home Ecosystem

Core Technologies



IoT Connectivity



Mobile Application



Cloud Platform

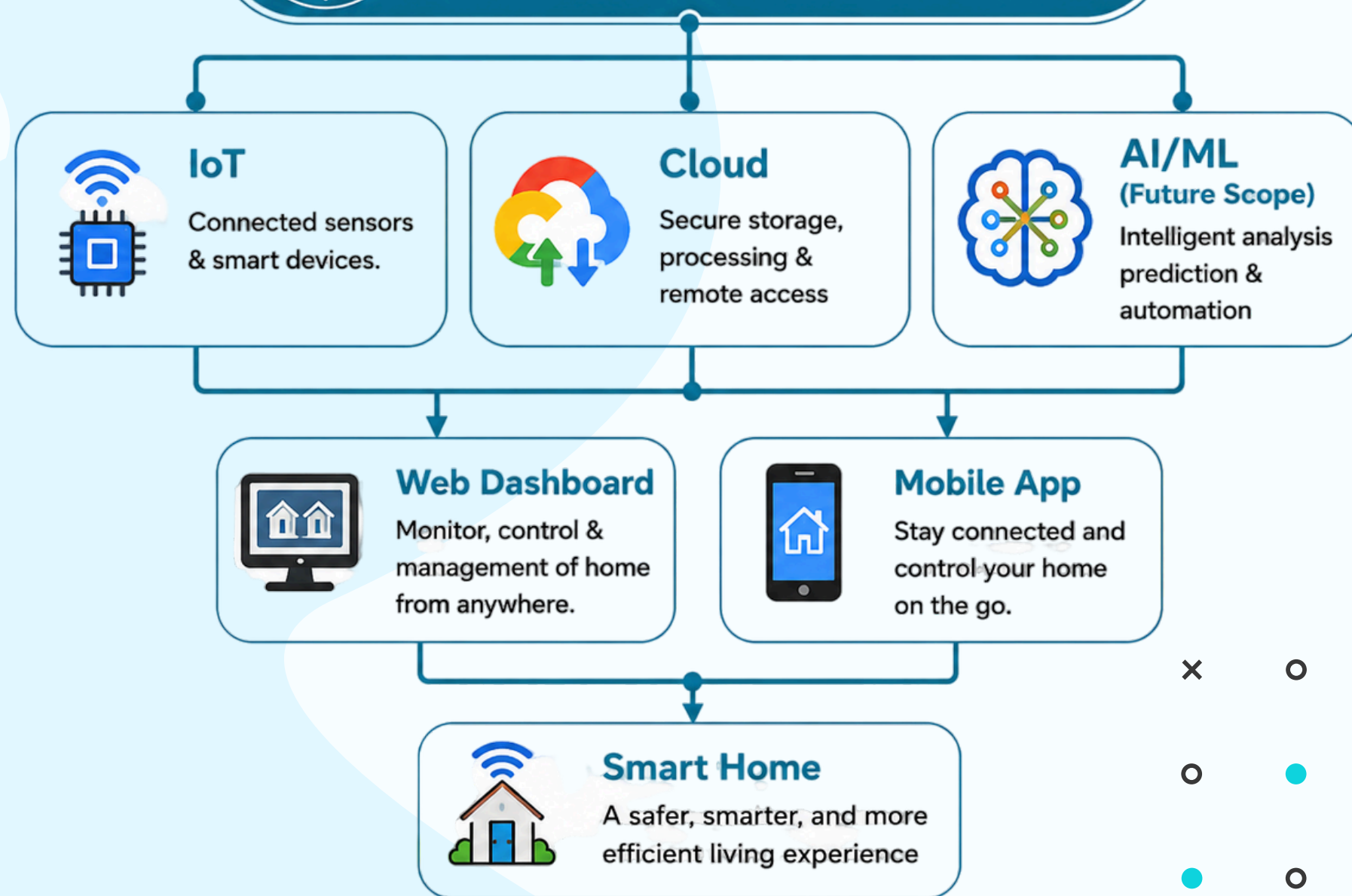


Web Dashboard

Secure cloud platform for real-time monitoring, remote access and future AI integration.



The Nexus Dome



Why The Nexus Dome?



Enhanced Safety



Anytime, Anywhere Control



Reduced Energy Consumption



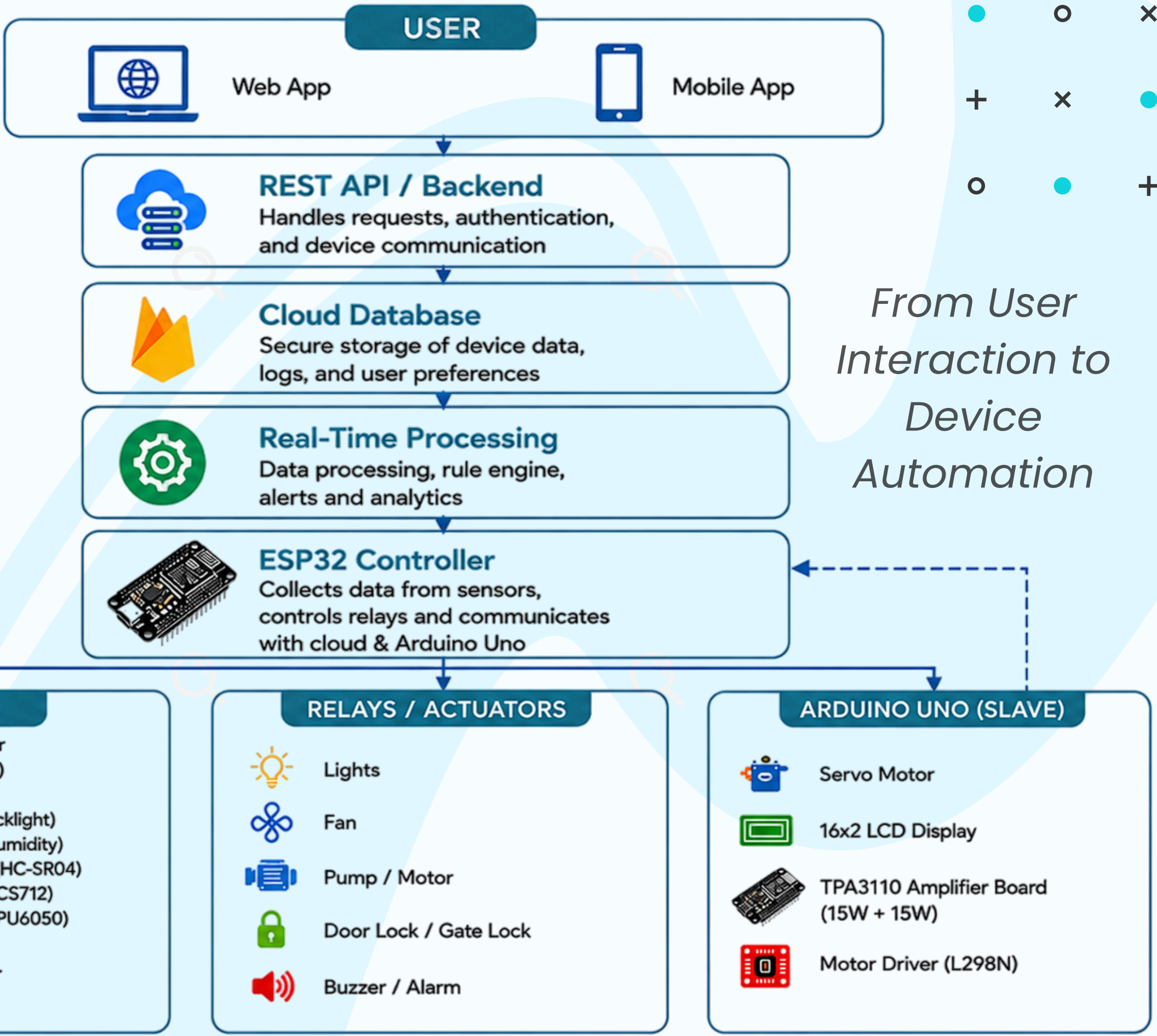
Intelligent Decision Support

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Architecture Highlights

- Centralized Device Management
- Real-Time Monitoring
- Secure Cloud Synchronization
- Remote Access & Control
- Modular & Scalable Design

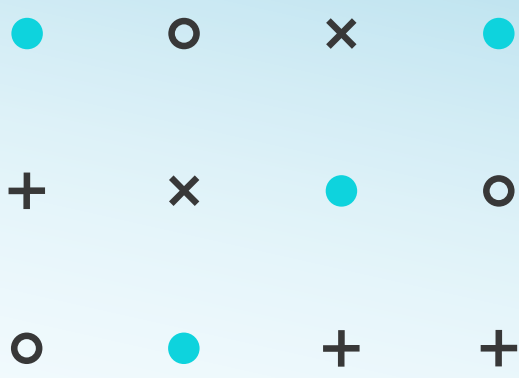
Scalable cloud architecture enabling secure communication, real-time monitoring, and intelligent home automation.



System Architecture

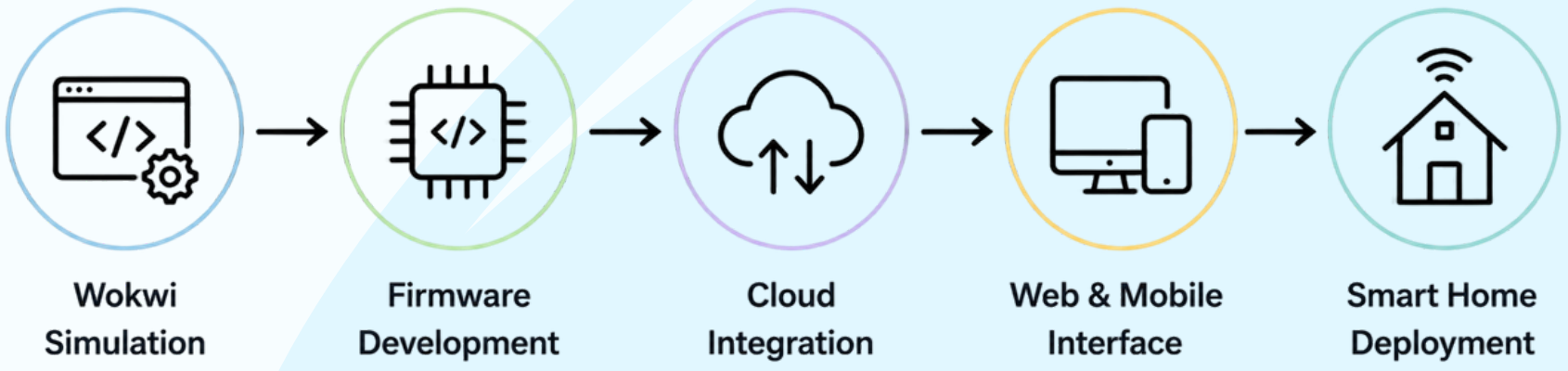
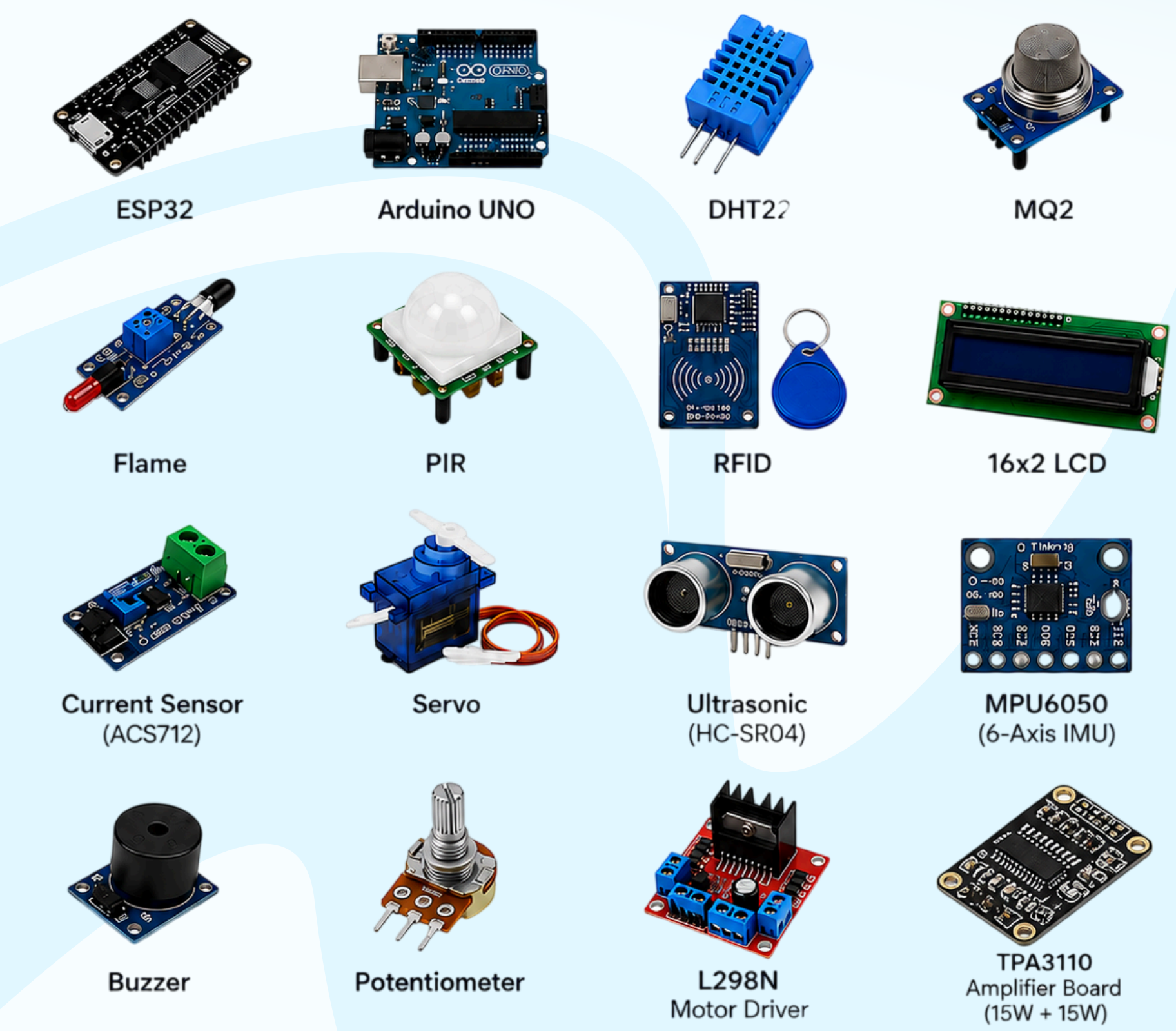
Real-time communication enables intelligent monitoring and automated decision-making.

Hardware & Software Stack



Software

Hardware



Development Workflow Overview

Innovation: Intelligent Web & Mobile Application

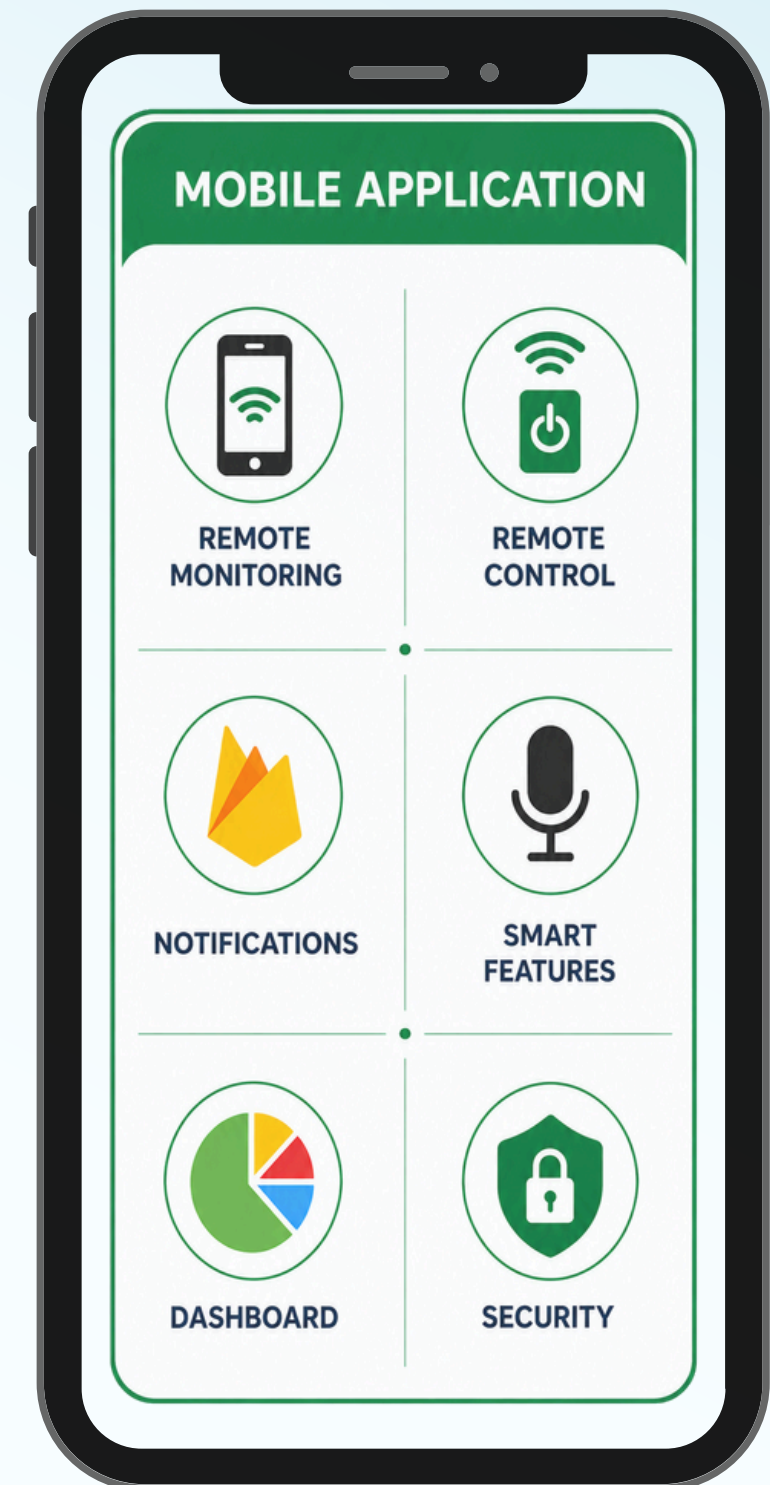


One Unified Platform to Monitor, Control, and Secure Your Entire Smart Home Anytime, Anywhere.

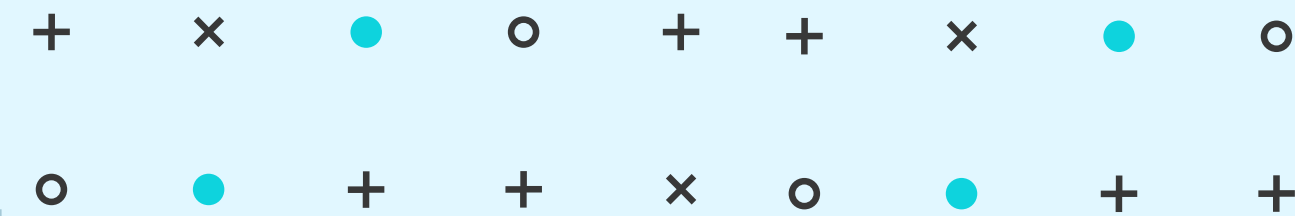
Key Innovations

- ☒ **Cloud Connected**
- ☒ **Real-Time Monitoring**
- ☒ **Remote Control**
- ☒ **Cross-Platform Access**
- ☒ **Emergency Notifications**
- ☒ **AI-Driven Automation**

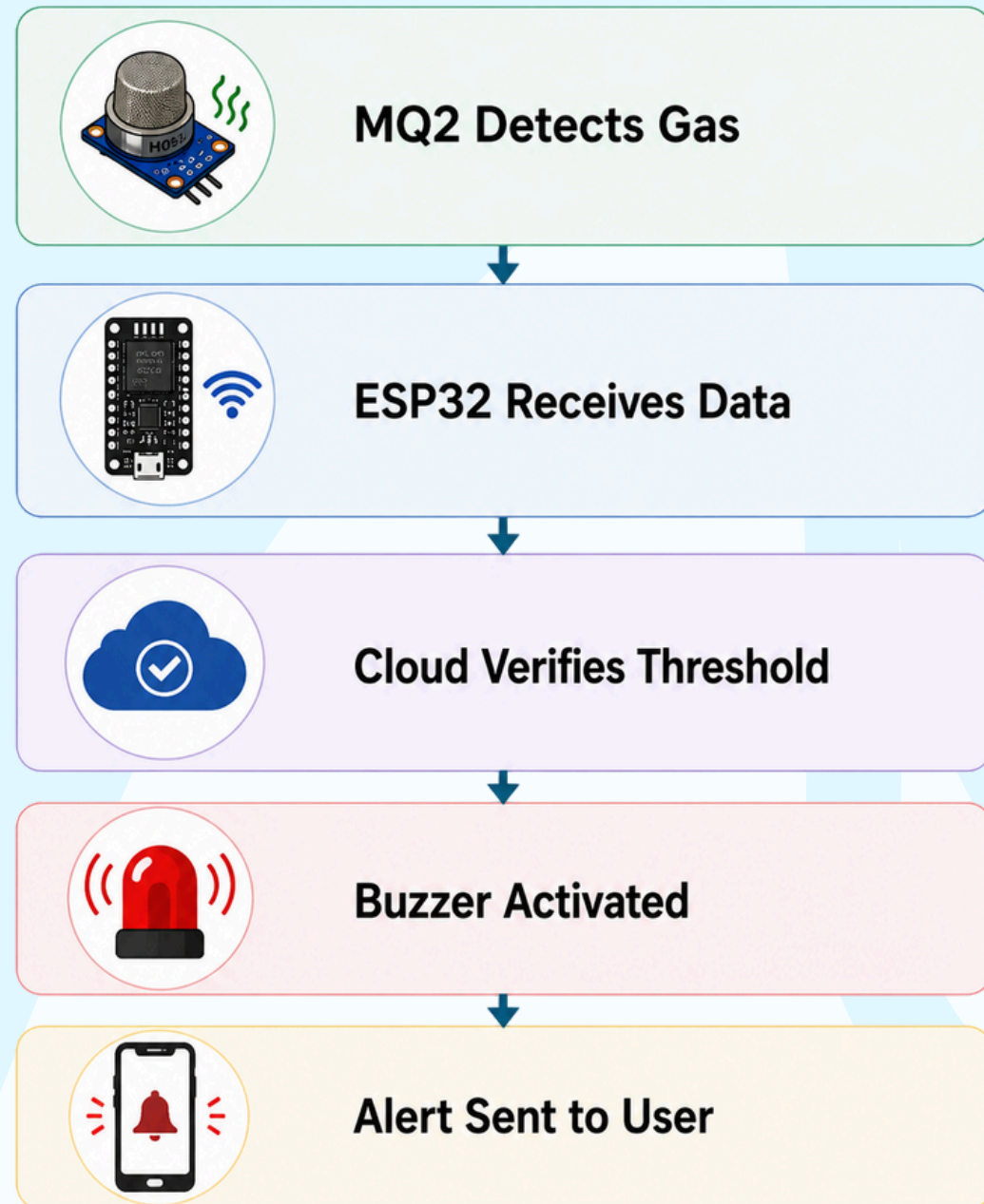
Monitor • Analyze • Automate • Protect



Live Automation Workflows

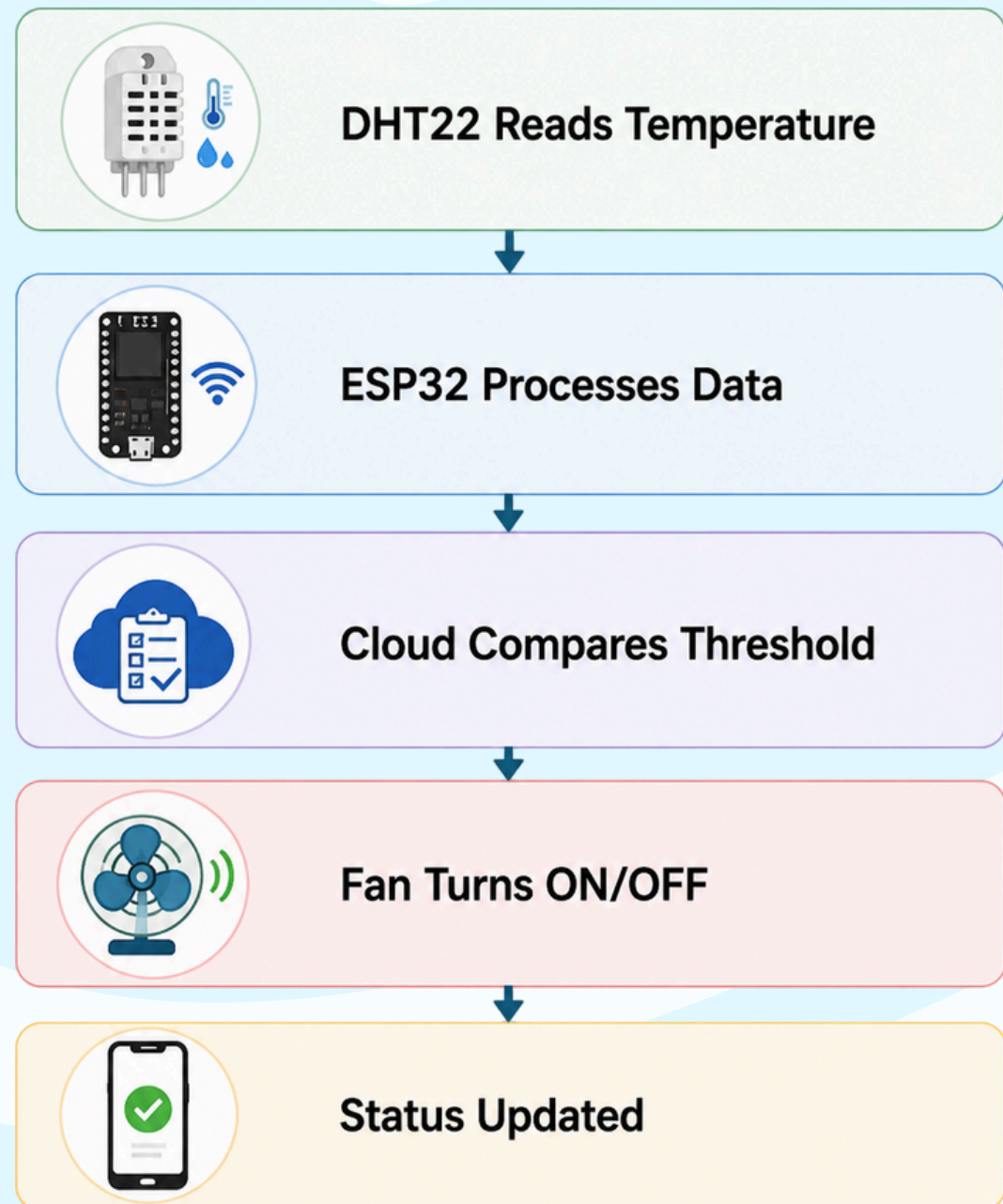


1. GAS LEAK DETECTION



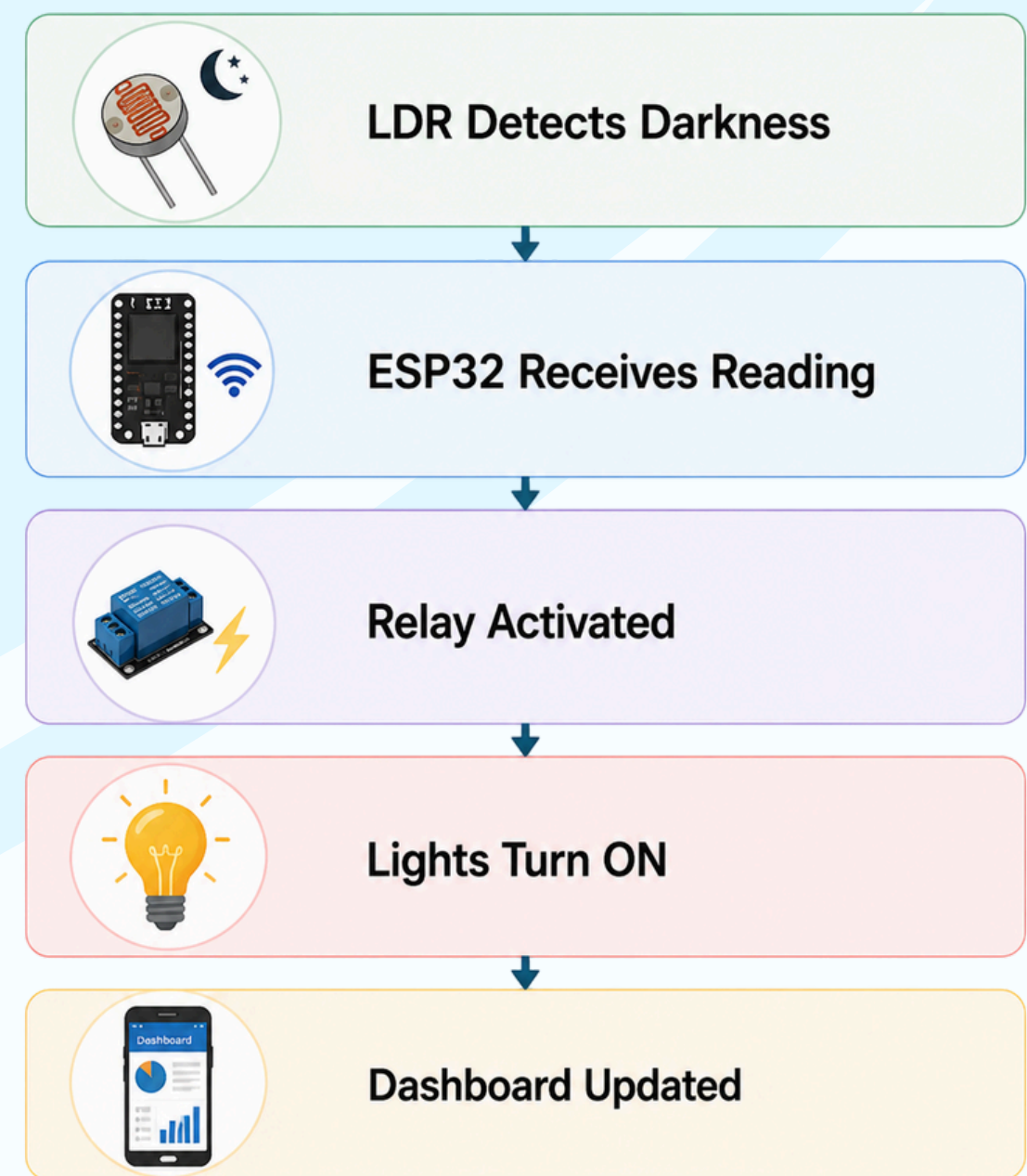
Scenario 1

2. SMART CLIMATE CONTROL



Scenario 2

3. AUTOMATIC LIGHTING



Scenario 3

Every automation follows a secure detection, processing, decision and response pipeline.

Live Automation Workflows

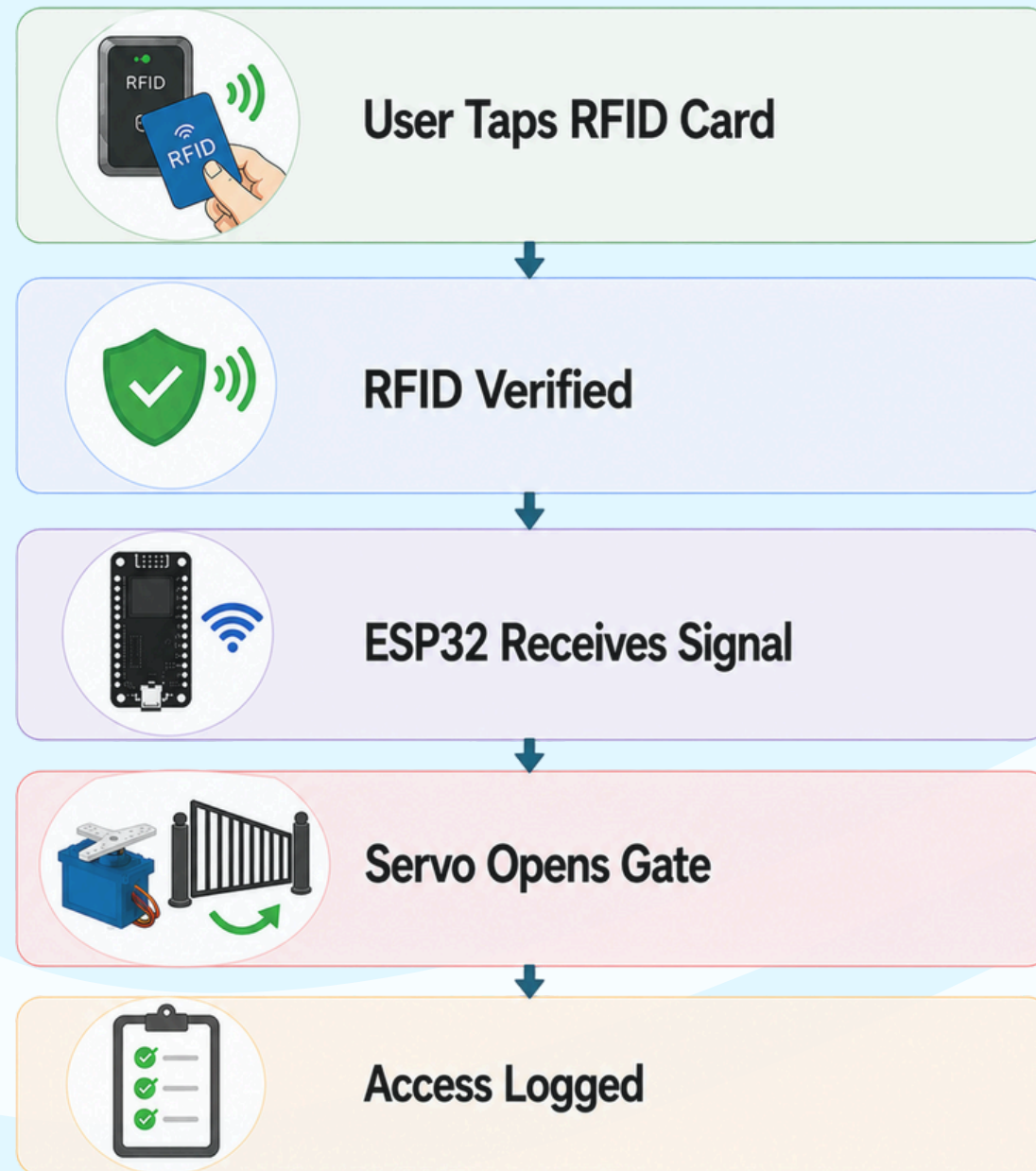
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4. MOTION DETECTION



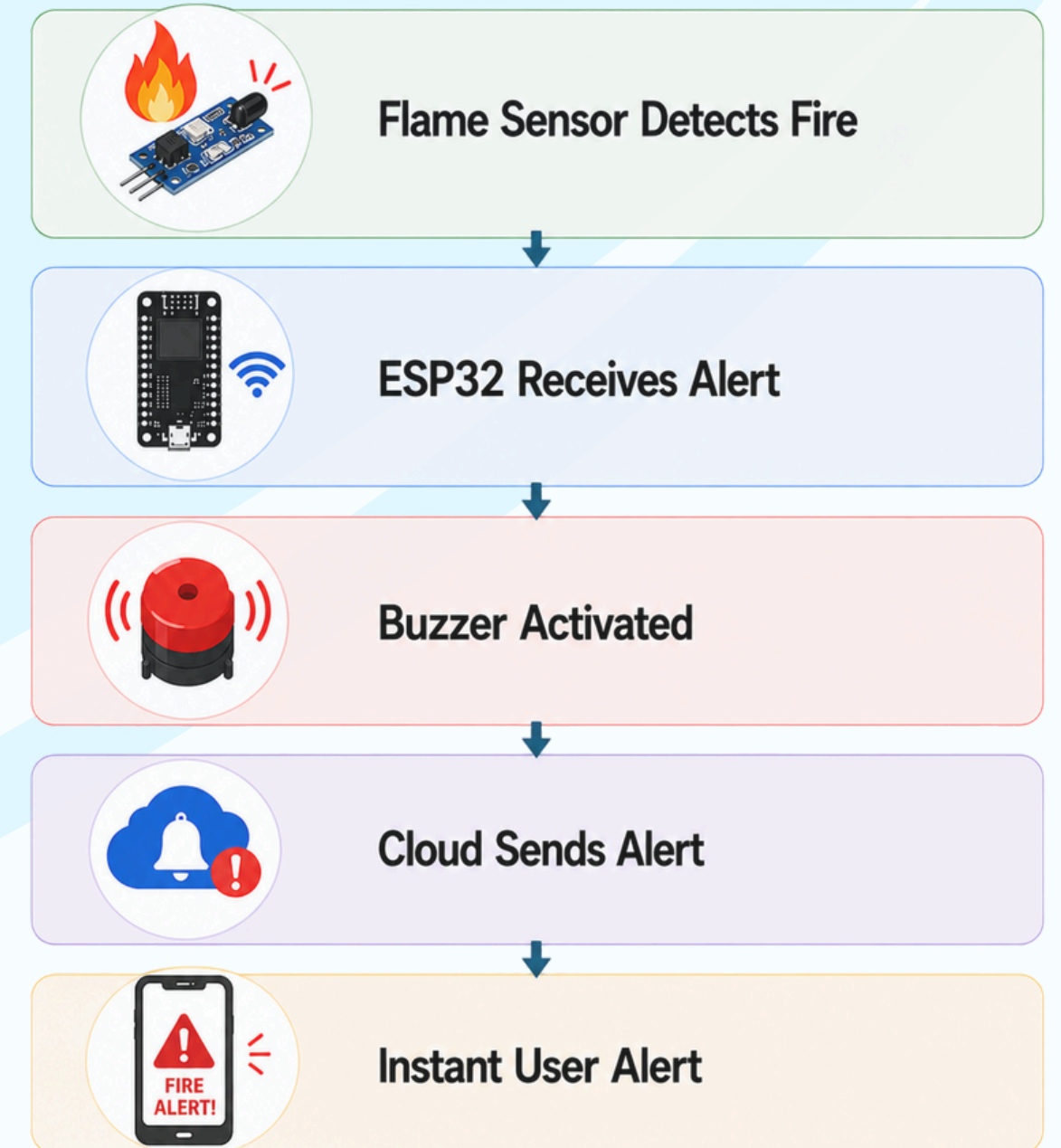
Scenario 4

5. RFID GATE ACCESS



Scenario 5

FIRE DETECTION ALERT



Scenario 6

Every automation follows a secure detection, processing, decision and response pipeline.

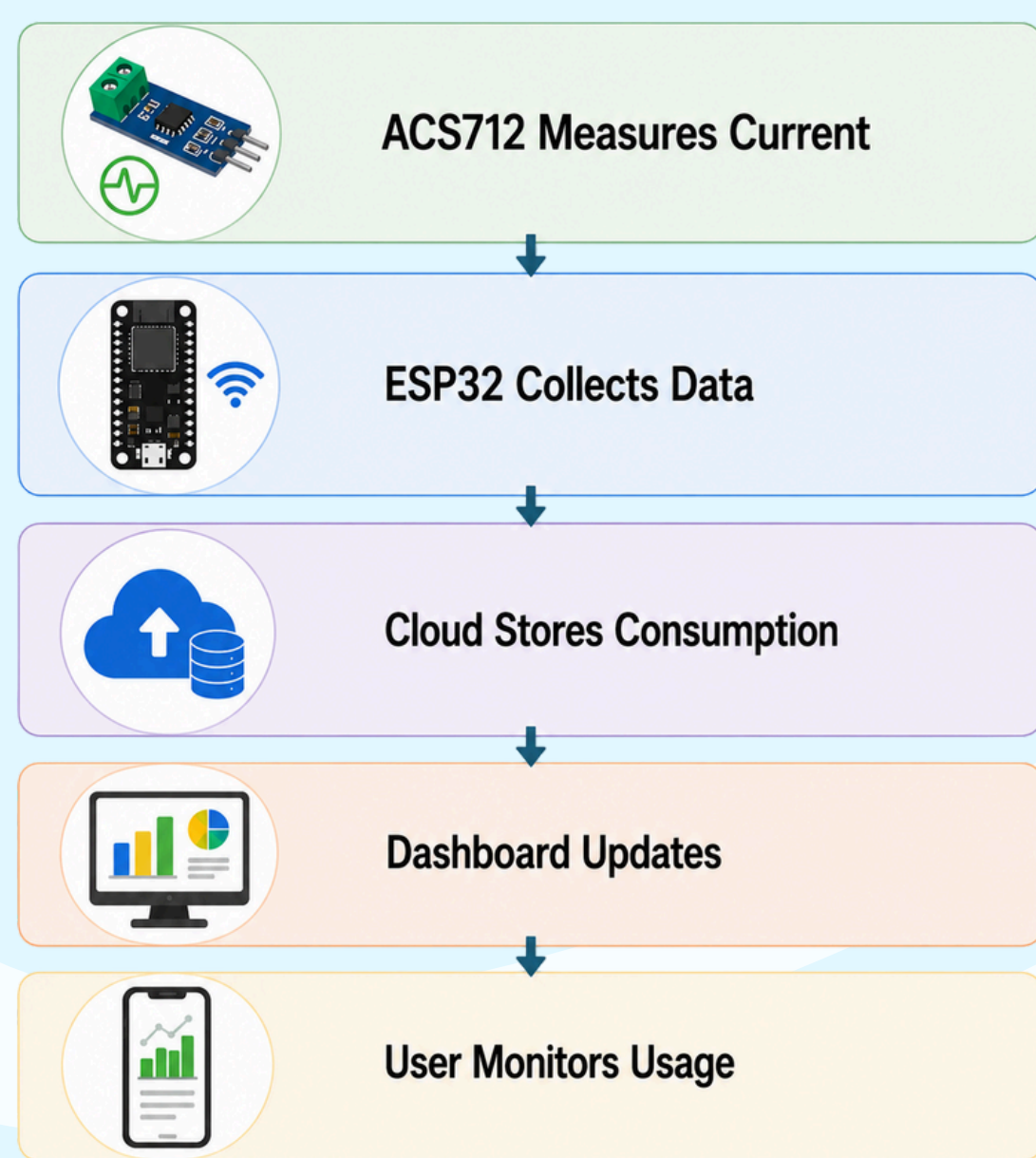
Live Automation Workflows

WATER TANK MONITORING



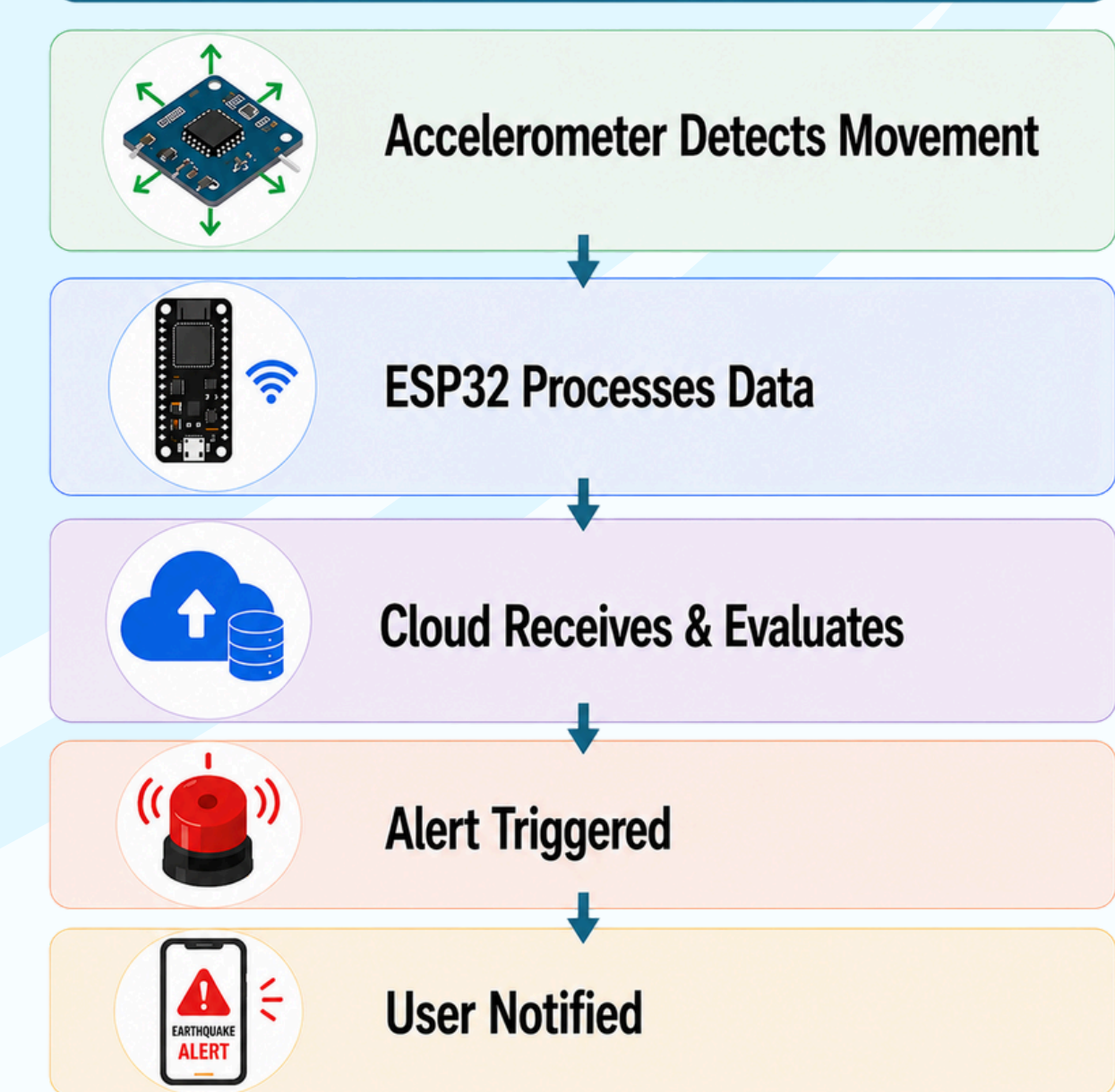
Scenario 7

ENERGY MONITORING



Scenario 8

EARTHQUAKE ALERT



Scenario 9

Every automation follows a secure detection, processing, decision and response pipeline.

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Competitive Advantages

Why Choose The Nexus Dome ?

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Key Advantages

- ✱

AI-Ready Architecture
Supports future AI-powered automation and smart decision-making.
- ✱

Low-Cost Deployment
Affordable hardware with open-source technologies.
- ✱

Cloud Connected
Remote monitoring and secure real-time synchronization.
- ✱

Multi-Platform Access
Control your smart home from web and mobile apps.
- ✱

Multi-Layer Safety
Monitors gas, fire, motion, energy and emergencies.
- ✱

Secure Smart Access
RFID-based authentication.

Cost Analysis

Component	Cost	Component	Cost
ESP32	350	PIR Motion Sensor	60
Arduino UNO	210	Flame Sensor	70
MQ2 Gas Sensor	150	Potentiometer	12
16×2 LCD Display	185	Ultrasonic Sensor (2)	300 (150 × 2)
DHT22 Sensor	115	ACS712 Current Sensor	125
RFID Module	250	MPU6050 Accelerometer	200
L298N Motor Driver	230	Servo Motor (2)	180 (90 × 2)
TPA3110 Amplifier Board	60	Buzzer (2)	28 (14 × 2)
		Wires, Connectors & Misc. Accessories	700



Total Prototype Cost (Approx.) ≈ ₹3,060

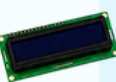
Deployment & Business Model

BUSINESS MODEL

1		Hardware Kit (Basic)	₹4,999 – ₹6,999 One-time
2		Hardware Kit (Advanced)	₹8,999 – ₹12,999 One-time
3		Cloud Subscription (Optional)	₹49 – ₹99 / month Recurring
4		Installation Service	₹500 – ₹1,500 One-time
5		Annual Maintenance (AMC)	₹500 – ₹1,000 / year Optional
6		Enterprise / Society Deployment	Custom quotation Project-based

PROTOTYPE COMPONENTS

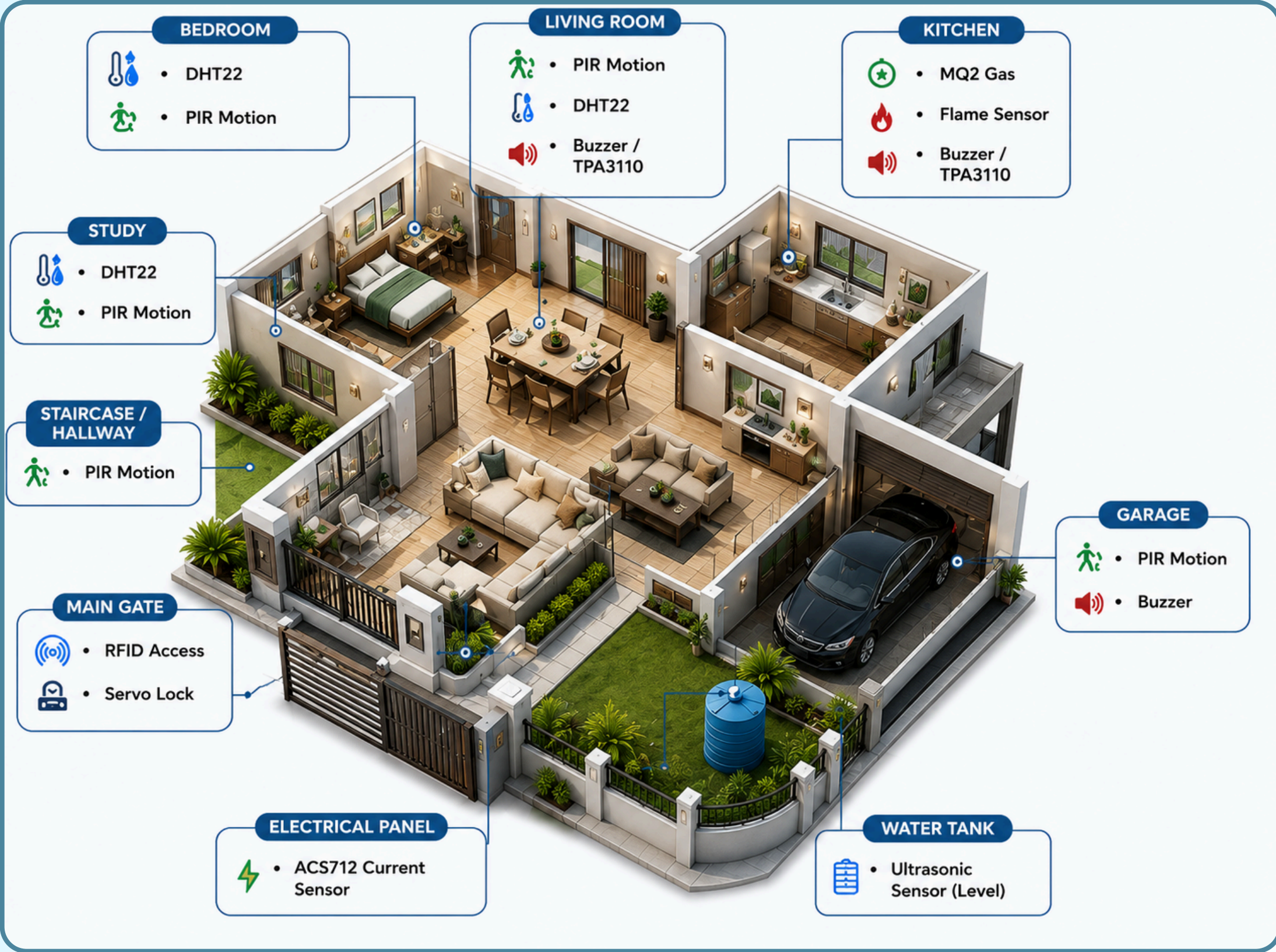
(Not Used in Real Home)

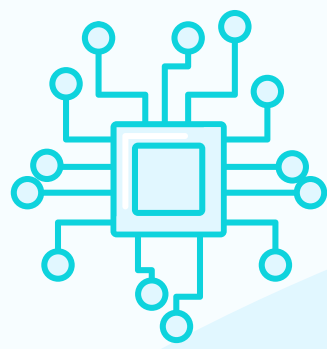
Component	
	Arduino UNO
	L298N Motor Driver
	Potentiometer
	16x2 LCD Display
	Servo Motor
	Buzzer

DEPLOYMENT-READY COMPONENTS

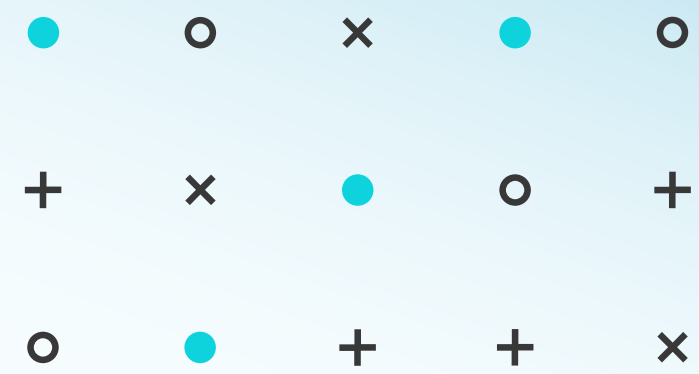
(Used in Real Smart Home)

	ESP32 DevKit
	MQ2 Gas Sensor
	DHT22 Sensor
	PIR Motion Sensor
	Flame Sensor
	+ 8 More Components



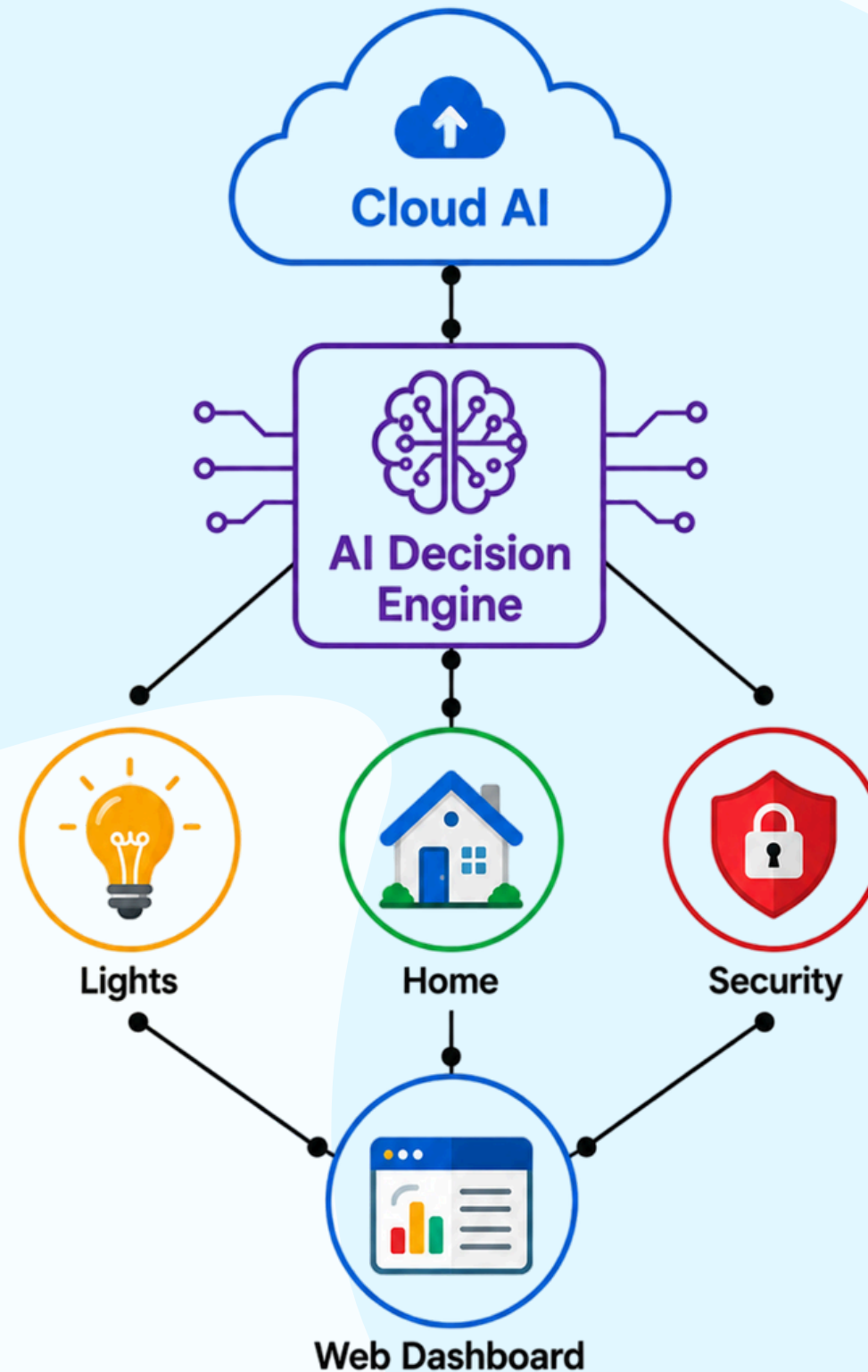


AI Features & Future Scopes



AI Features (Current)

- ✓ **Real-Time Alert System**
- ✓ **Cloud Data Analytics**
- ✓ **Intelligent Automation Rules**
- ✓ **Energy Optimization**
- ✓ **Occupancy Detection**
- ✓ **Remote Device Control**



Turning Real-Time Sensor Data
into Intelligent Decisions

Future Scope

- ⚙️ **AI Behaviour Learning**
- ⚙️ **Predictive Maintenance**
- ⚙️ **Digital Twin**
- ⚙️ **Computer Vision Security**
- ⚙️ **Voice Assistant Integration**
- ⚙️ **Solar & Smart Grid Integration**

AI-Powered Smart
Home Automation
Ecosystem

Redefining
Smart Living
Through AI & IoT

THE NEXUS DOME

Thank You

"Building Smarter
Homes, One
Innovation at a Time."

"Turning Ideas into
Intelligent Living."



Questions & Discussion
We'd be happy to answer your questions.



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